SEPTEMBER REVENUE · SAHYADRI STORES

~~Rs 24,03,00,20~~

Our AI agent said this, with full confidence.

The real answer is **Rs 2,40,300**

No error. No warning. Just a wrong number, a hundred times too big.

THE PROBLEM

AI agents fail in three ways. Only one makes noise.

Loud

The tool throws an error

```
no such column: amount
```

Easy to catch, easy to retry.

Quiet

The number is just wrong

```
prices read in paise, a
missing GROUP BY
```

The query runs fine. Nobody notices.

Spreading

The next answer builds on it

```
turn 2 reuses turn 1's number
```

One mistake becomes many.

PBA 1 catches the loud ones. **PBA 2** catches all three.

QueryMend

Ask your shop database a question in plain English. If the answer is wrong, it notices, fixes it, or tells you it can't.

Executor
writes the SQL

Monitor
checks everything

Recovery
fixes or stops

A person
approves personal data

qwen2.5:3b · LangGraph · SQLite · Chroma · fully offline · free

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Topic 17: an agent that detects and recovers from its own tool failures

PBA 1 · UNITS 1 AND 2

One agent that notices when its tool fails

Diagram goes here

`report/diagrams/drawio/01_pba1_agent.png`

THE SLIDE VERSION

One generic message for every error

Sent the same broken query 7 times. Told a user to run a DELETE by hand.

OUR REFINEMENT

Sort the error, then act on it

A specific hint, at most 3 tries, and the tool is locked when retrying can't help.

Blind spot: wrong data that raises no error goes straight to the user.

PBA 2 · UNITS 3 AND 4

Three agents, one job each

Diagram goes here

`report/diagrams/drawio/02_pba2_team.png`

Executor

Writes and runs the SQL. The only agent that touches the data.

Monitor

Plain rules, no LLM. Checks every result and every answer.

Recovery

Advice from our error guide, or a safe stop after 3 retries.

A REAL RUN

One question, two silent mistakes, both caught

Diagram goes here

report/diagrams/drawio/05_one_recovery.png

1 **Ambiguous column**

The tool errors. Recovery pulls advice from the guide.

2 **Missing GROUP BY**

No error at all. The Monitor reads the SQL and quarantines it.

3 **Corrected query**

5 rows. The answer matches the data.

traces/ta2/failures.txt

It remembers, it resumes, it reads its own guide

Diagram goes here

report/diagrams/drawio/03_memory_rag.png

55 → 63 — "And in September?" works with a checkpointer, and fails without one.

Restart — "Premium product = above Rs 1500" is recalled in a brand-new process.

k = 3 — A two-part question gets both parts. "Connect to MySQL?" gets "I don't know".

Logged — Every failure, its SQL and what Recovery decided, in SQLite.

ERROR PROPAGATION · 10 RUNS EACH

With the Monitor on, a bad number stops spreading

Diagram goes here
report/diagrams/drawio/04_containment.png

10 / 10

Monitor off: the wrong number was carried into the follow-up answer.

0 / 10

Monitor on: quarantined before any agent could read it.

Cost: 2 extra model calls, about 2 seconds.

SAME 5 CASES · PBA 1 VS PBA 2 · 3 RUNS EACH

Fewer wrong answers, more model calls. Both are true.

GOOD NEWS

4 → 0

wrong answers given to the user, out of 15 runs

0 → 3

silent corruptions caught, out of 3

THE PRICE

2 → 6

model calls when the data is corrupt

+1

model call even on a clean question

FOUND BY OUR OWN TESTS

A missing GROUP BY slipped past every number check. The Monitor now reads the SQL too. Clean question: 3/3 wrong became 3/3 correct.

LIVE DEMO

Four runs, about three minutes

1. **One corrupted result, Monitor off, then on**

   `python ta2/run.py containment`

2. **Five injected failure types and the evidence table**

   `python ta2/run.py failures`

3. **Personal data paused for a person: approve, then reject**

   `python ta2/run.py privacy`

4. **Your question, or your edge case**

   every run is also saved in traces/ as a backup

WHAT'S NEXT

From a class project to a tool a shop owner can trust

Owners set their own limits

"An order is never above Rs 50,000"

Real databases, read-only

Postgres or MySQL through a read-only user, same checks

Show the work

The answer, the SQL behind it, and a note whenever the Monitor stepped in

Thank you. Questions?

github.com/Swaraj-Mandre/FailureDetectionAgent